LITERATURE SURVEY

1. Review of Key Research Papers

1.1 Xerox Instant Print Kiosk

The Xerox Instant Print Kiosk is a fully automated, self-service document handling system designed to enable users to independently print, copy, scan, and fax documents in public or semi-public environments. The system eliminates the need for staff assistance by integrating touchscreen navigation, cloud document access, mobile printing support, and secure user authentication. It supports walk-in users by allowing them to upload documents via USB, email, or mobile applications, and complete transactions using digital payment methods. The kiosk is intended for deployment in hotels, universities, airports, libraries, and coworking spaces where fast and convenient printing services are required. The system provides features such as automated billing, usage tracking, encrypted document processing, and remote device management to ensure operational efficiency. While this model offers a robust commercial solution, it remains costly for small businesses and lacks customization flexibility for specific regional requirements. Overall, the Xerox Instant Print Kiosk represents a mature and reliable smart printing solution tailored for high-traffic commercial environments.

1.2 Xerox Workplace Kiosk App

The Xerox Workplace Kiosk App transforms compatible Xerox printers into multifunctional self-service stations that enable users to access print, copy, scan, and email functionalities through mobile-based authentication. The system integrates cloud management features, QR-code login, and support for Google Drive, OneDrive, Dropbox, and other cloud storage services. It operates on a pay-per-use model, allowing organizations to deploy self-managed kiosks without dedicated operators. Payment processing, account management, and device monitoring are centralized through a cloud dashboard, enabling administrators to track usage, security, and billing. The kiosk app is ideal for universities, hostels, coworking spaces, and small businesses seeking a low-cost self-service printing solution. Although the app provides strong security and mobility features, it is limited to specific Xerox printer models and requires stable network connectivity. Overall, the Workplace Kiosk App delivers a flexible, software-driven approach to smart, automated document services.

1.3 Canon UniFLOW Print Kiosk

Canon UniFLOW Print Kiosks represent a robust and secure self-service printing solution designed for organizations, enterprises, educational institutions, and public spaces requiring controlled and efficient document management. The UniFLOW platform integrates print, copy, scan, and cloud-based workflows into a unified environment that enhances user convenience while maintaining strict document security. Users can authenticate using ID cards, PIN codes, or mobile-based QR login, enabling secure print release and preventing unauthorized access to sensitive documents. The system provides flexible printing options through cloud services, email, USB drives, and local network submission. Built-in cost tracking, user quotas, and detailed administrative analytics support efficient resource management and cost control. Additionally, UniFLOW automates supply monitoring, print job routing, and device usage optimization to improve operational efficiency. While the system offers advanced security and scalability, its dependency on enterprise-level infrastructure and relatively high setup cost may limit accessibility for smaller organizations. Overall, Canon UniFLOW Print Kiosks deliver a highly secure, customizable, and integrated approach to modern self-service printing.

1.4 PrintWithMe Self-Service Printing Network

PrintWithMe is a widely adopted, decentralized self-service printing network designed to provide convenient and secure document printing in public and semi-public spaces such as cafés, residential buildings, coworking hubs, hotels, and community centers across the United States. The system enables users to upload documents through email or the PrintWithMe web portal, after which the print job is processed via encrypted, temporary cloud storage to ensure privacy. Customers complete payment online and retrieve prints directly from onsite PrintWithMe-enabled printers without requiring staff assistance. The platform integrates remote monitoring, automated supply management, and predictive maintenance, enabling high uptime and reduced operational burden for host locations. Performance evaluations in deployed environments highlight significant improvements in accessibility and user convenience, particularly for individuals without personal printers or those requiring urgent printing.

1.5 Blinkit Print-on-Demand Service

The rapid expansion of quick commerce (Q-commerce) in India has led to novel service offerings beyond traditional grocery delivery, including on-demand print services. In 2022, the Zomato-owned Blinkit introduced a feature allowing users to upload documents via the Blinkit app and receive printed copies at their doorstep within minutes, representing a convergence of digital convenience and ofline document handling. This case study explores consumer behavior, privacy concerns, and adoption likelihood related to Blinkit's print-on-demand feature. Data from industry reports indicates that only around 14 % of Indian consumers expressed willingness to use quick delivery platforms for printing needs, citing convenience and delivery speed as primary motivators, while privacy concerns and habituation to existing printing methods remain key barriers. Operational challenges such as maintaining confidentiality and timely delivery impact service adoption and sustainability. These findings provide valuable insights into expanding Q-commerce offerings into document services and underscore the need for enhanced user trust and improved logistics to increase long-term adoption. *(Data on Blinkit's service launch is derived from news reports of August-September 2022.)*

LITERATURE REVIEW

Existing smart printing systems such as PrintWithMe are designed for general public usage, not for academic institutions, which creates several limitations in a campus environment. These systems often have higher per-page costs, making them unsuitable for students who require frequent and bulk printing for assignments, lab manuals, and study materials. They rely heavily on internet connectivity and cloud uploads, causing delays during peak network usage. Most existing services lack student-centric features like department-wise printing templates, faculty-priority access, or integration with ERP portals and classroom platforms such as Moodle. Additionally, they are not built for high printing volume within short time windows, such as during exams or submissions. Their focus on commercial use means limited customization, no offline printing option, and no provision of preloaded academic materials or course-specific document folders.

Smart xerox service project is purpose-built for students and faculty, making it far more efficient, affordable, and relevant than generic smart printing services. By using a local network, it ensures instant document transfer and reduces dependence on internet speed. The system provides low-cost, campus-friendly printing options that match student budgets. It supports features like secure login, fast document release, department-wise material categories, faculty priority mode, and direct integration with college ERP platforms for seamless printing of assignments and notices. Smart xerox service can also store frequently used academic materials (timetables, syllabus, manuals, forms), enabling one-tap printing. Additionally, the system saves a significant amount of time by eliminating long queues, allowing students to upload documents remotely and collect prints instantly without waiting. The integrated cashless payment options such as UPI, card, or wallet ensure quick and hassle-free transactions, further improving the user experience. Because it is customized exclusively for a campus environment, it offers better speed, convenience, privacy, ease of use, and academic alignment, making it significantly superior to existing public printing solutions.